

Conformal False Discovery Rate Control for Molecular Property Prediction Under Covariate Shift

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Abstract

Virtual screening requires selecting a subset of compounds predicted to satisfy a property threshold (e.g., sufficient solubility, permeability, or metabolic stability) from a much larger, unlabelled library. Because thousands of compounds are tested simultaneously, naive thresholding of a regression or classification score inflates the false discovery rate (FDR) among the selected set, and the miscalibration is compounded when the model is applied to compounds that are structurally dissimilar from the calibration data — the norm rather than the exception in prospective screening. We introduce a conformal hypothesis-testing framework for this setting: for each candidate compound we test the null hypothesis that its property value does not exceed a decision threshold τ , and compute an exact, finite-sample-valid p-value against an empirical null distribution formed from the negative (sub-threshold) subset of an independent calibration set. To adjust for distribution shift, we further introduce three importance-weighting schemes — entropy balancing, direct Kolmogorov–Smirnov (KS) distance minimisation, and local Tanimoto-similarity (k-nearest-neighbour) weighting — that re-weight the calibration set so that its Morgan-fingerprint neighbour-distance distribution matches that of the target (test) compounds, restoring approximate exchangeability under scaffold- or cluster-induced covariate shift. We evaluate the framework on real ADMET endpoints (Caco-2 permeability, human liver microsomal intrinsic clearance, and kinetic solubility) under random, cluster-based, and scaffold-based splits, on a 100-assay ChEMBL bioactivity benchmark, and on a synthetic scaffold R-group enumeration designed to isolate covariate shift from all other confounders. Unweighted conformal FDR control is reliable under random splits but degrades substantially under structural splits, with median true FDR exceeding the nominal level for the majority of assays under cluster-based splitting. Importance weighting reduces this violation, at the cost of a smaller, more conservative selected set when the shift is severe.

1 Introduction

Machine learning models for molecular property prediction — solubility, permeability, metabolic stability, target potency — are typically deployed to screen tens of thousands to millions of virtual compounds and return a shortlist for synthesis. This is fundamentally a multiple-hypothesis-testing problem: each compound in the library corresponds to one hypothesis test of whether its true property value crosses a decision threshold, and the fraction of false positives in the returned shortlist (the false discovery rate, FDR) is the quantity a practitioner actually cares about, not the pointwise accuracy of the underlying regression or classification model. Reporting a raw model score, or even a well-calibrated probability, does not by itself control the FDR of a shortlist selected at some cutoff, because that guarantee requires accounting for the number of hypotheses tested and the correlation structure between them [Benjamini and Hochberg, 1995].

Conformal prediction offers a route to distribution-free, finite-sample-valid p-values without parametric assumptions on the model’s error distribution [Vovk et al., 2005, Shafer and Vovk, 2008], and recent work has shown how to turn conformal p-values into FDR-controlling procedures via the Benjamini–Hochberg (BH) procedure, both for outlier/novelty detection [Bates et al., 2023] and for selection problems more broadly [Jin and Candès, 2023]. All of these guarantees, however, rest on an exchangeability assumption between the calibration data and the test point. In molecular property prediction this assumption is routinely violated by design: the very purpose of a model is to extrapolate to new chemical series, and reliable retrospective evaluation of a model in drug discovery requires scaffold- or cluster-based splitting precisely because random splits over-estimate performance on future, structurally novel compounds [Martin et al., 2017, Schiebroek et al., 2026]. A calibration set built from already-synthesized chemical matter is therefore systematically closer to itself, in fingerprint space, than to the virtual library a model will actually be asked to triage — a covariate shift that can silently break the validity of a naively BH-adjusted conformal p-value.

This paper develops and empirically stress-tests a conformal FDR-control framework for this setting. Concretely, we:

1. formalise molecular threshold screening as a one-sided hypothesis test per compound and define a *Subpopulation Prediction Distribution Test* (SPDT) that calibrates against the model’s predictions on the sub-threshold subpopulation only, rather than against a global residual distribution (Section 3);
2. extend this test with three importance-weighting schemes — entropy balancing, KS-distance minimisation, and local Tanimoto k-NN weighting — that correct for covariate shift between calibration and target compounds by re-weighting the calibration set in fingerprint neighbour-distance space (Section 3.2);
3. quantify, on three ADMET endpoints, a 100-assay ChEMBL bioactivity benchmark, and a controlled synthetic scaffold-swap experiment, how badly naive (unweighted) conformal FDR control is violated under scaffold- and cluster-based splits, and to what extent importance weighting restores nominal control (Section 4–5).

Scope of this draft. This document is an initial, code-grounded draft assembled directly from the accompanying implementation (`src/conformal_predictor.py`, `src/train_xgboost.py`, `src/train_xgboost_schiebroek.py`, `src/scaffold_experiment_1.py`) and the result artefacts already produced (`results/*.csv`, `results/*.pkl`). Numbers reported in Section 5 are computed from those artefacts; sections marked [TODO:] indicate analyses the code supports but for which results have not yet been generated or written up.

2 Related Work

Conformal prediction and weighted exchangeability. Conformal prediction [Vovk et al., 2005] converts any model score into finite-sample-valid p-values or prediction sets under the sole assumption of exchangeability between calibration and test data. Tibshirani et al. [2019] extend the framework to *covariate shift*, showing that replacing uniform calibration weights with likelihood-ratio importance weights restores validity when the test covariate distribution differs from the calibration distribution but the two share a common conditional law of $Y \mid X$. Our weighting schemes are practical, density-ratio-free instantiations of this idea, using nearest-neighbour Tanimoto distance in fingerprint space as a proxy for the unknown likelihood ratio.

Conformal p-values, outlier testing, and FDR control. Bates et al. [2023] introduce conformal p-values for testing whether a new point is an outlier relative to an inlier calibration set, and show that Benjamini–Hochberg applied to these p-values controls the FDR among flagged outliers. Our setting differs in target and structure (Section 3.1.5): we test a directed hypothesis about a continuous, *labelled* response variable rather than performing unsupervised out-of-distribution detection, and our empirical null is built from the model’s own predictions on a labelled sub-threshold subpopulation rather than from a generic conformity score over the feature space. The underlying multiple-testing machinery — Benjamini–Hochberg [Benjamini and Hochberg, 1995], Benjamini–Yekutieli for arbitrary dependence [Benjamini and Yekutieli, 2001], and the Holm and Hochberg step-down/step-up procedures for family-wise error control [Holm, 1979, Hochberg, 1988] — is standard and implemented directly (`ConformalPredictor.p_adjust`).

Risk-controlling prediction and selective inference. Risk-controlling prediction sets (RCPS; Bates et al., 2021) calibrate a single global threshold so that a user-specified risk (e.g., expected loss) is controlled in expectation over the population; the closely related conformal risk control framework [Angelopoulos et al., 2023] generalises this to a broad class of monotone risks. These frameworks calibrate a *global* error distribution $|Y - \hat{Y}|$ over the full population, which can be mis-calibrated under heteroskedastic error or population-dependent bias. SPDT instead conditions calibration on the sub-threshold subpopulation specifically, trading generality for a null distribution that mirrors the model’s actual behaviour on definitionally-negative compounds.

Distribution shift in molecular machine learning. It is well established that random train/test splits over-estimate prospective model performance in drug discovery, and that scaffold- or cluster-based splits are a more faithful proxy for deployment on novel chemical series [Martin et al., 2017, Wu et al., 2018]. We adopt the cluster-based (Butina) splitting protocol of Schiebroek et al. [2026] for the ChEMBL bioactivity benchmark (Section 4.3) and use the resulting benchmark as the primary evidence that covariate shift, not merely model miscalibration, is responsible for FDR violations under structural splits.

3 Methods

3.1 Problem formulation and the Subpopulation Prediction Distribution Test (SPDT)

We consider a setting where each molecule i is characterised by a feature vector $X_i \in \mathcal{X}$ (a Morgan fingerprint, optionally concatenated with RDKit physicochemical descriptors) and a target property Y_i , which may be continuous (e.g., $\log P$, intrinsic clearance) or binary (e.g., an assay-specific active/inactive call). Let $f : \mathcal{X} \rightarrow \mathbb{R}$ be a model trained to predict Y (or, in the classification case, the probability of the positive class), yielding scores $\hat{Y}_i = f(X_i)$. The screening objective is to act as an asymmetric filter: for a novel, unlabelled compound with features X_{new} and unknown true property Y_{new} , we test the null hypothesis

$$H_0 : Y_{\text{new}} \leq \tau,$$

where τ is a pre-specified decision threshold (e.g., $\tau = 200 \mu\text{M}$ for kinetic solubility, or $\tau = 0.5$ on the predicted probability scale for a classification endpoint). Rejecting H_0 corresponds to nominating the compound for synthesis or follow-up.

To evaluate H_0 without parametric assumptions on the model’s error distribution, we use a **Subpopulation Prediction Distribution Test (SPDT)**. Unlike standard conformal regression or risk-controlling prediction sets [Bates et al., 2021], which calibrate a global residual distribution $|Y - \hat{Y}|$ over the whole calibration set, SPDT evaluates the raw prediction $\hat{Y}_{\text{new}}$ against an empirical null distribution built exclusively from the sub-threshold (definitionally negative) subpopulation.

3.1.1 Calibration and empirical null distribution

Let $\mathcal{D}_{\text{cal}} = \{(X_i, Y_i)\}_{i=1}^n$ be a calibration set independent of the training data (in our experiments, a held-out validation split). We construct the conditioned calibration subset

$$\mathcal{D}_{\text{cal}}^{\leq \tau} = \{(X_i, Y_i) \in \mathcal{D}_{\text{cal}} : Y_i \leq \tau\}, \quad m = |\mathcal{D}_{\text{cal}}^{\leq \tau}|,$$

and extract the corresponding model predictions to form the empirical null pool

$$\mathcal{P}_0 = \{\hat{Y}_i : (X_i, Y_i) \in \mathcal{D}_{\text{cal}}^{\leq \tau}\}.$$

By conditioning on $Y_i \leq \tau$, $\mathcal{P}_0$ captures both the intrinsic variance of the sub-threshold subpopulation and any systematic model bias specific to that regime, sidestepping the need to model heteroskedasticity across the full response range. We define the empirical CDF of $\mathcal{P}_0$ as

$$F_m(t) = \frac{1}{m} \sum_{\hat{Y}_i \in \mathcal{P}_0} \mathbb{I}(\hat{Y}_i \leq t).$$

3.1.2 Conformal p-value and decision rule

Given a new compound with prediction $\hat{Y}_{\text{new}} = f(X_{\text{new}})$, and assuming exchangeability between $\hat{Y}_{\text{new}}$ and $\mathcal{P}_0$ under H_0 , the (uniformly weighted) conformal p-value is

$$p(X_{\text{new}}) = \frac{1 + \sum_{\hat{Y}_i \in \mathcal{P}_0} \mathbb{I}(\hat{Y}_i \geq \hat{Y}_{\text{new}})}{m + 1}. \quad (1)$$

The +1 in numerator and denominator guarantees exact finite-sample validity: under H_0 and exchangeability, $p(X_{\text{new}})$ is super-uniform on $[0, 1]$. We reject H_0 at level α if $p(X_{\text{new}}) \leq \alpha$. The implementation (`ConformalPredictor.predict_pvalues`) generalises this to importance-weighted calibration samples w_i and an optional test weight w_{test} , following Tibshirani et al. [2019]:

$$p_j = \frac{\sum_i w_i \mathbb{I}(\alpha_i \geq \alpha_j) + w_{\text{test},j}}{\sum_i w_i + w_{\text{test},j}},$$

where α denotes either the raw model score (SPDT, as above) or a nonconformity score $Y_i - \hat{Y}_i$ computed over the *full* calibration set (Section 3.1.3); the indicator direction flips accordingly. Uniform weights ($w_i \equiv 1$) recover Eq. (1) exactly.

3.1.3 Two ways to pool calibration information

The implementation supports two calibration pools, both computed for every dataset in Section 5:

- **Score-based, negatives-only** (SPDT as defined above): calibration is restricted to $\mathcal{D}_{\text{cal}}^{\leq \tau}$ and the conformity statistic is the raw model score $\hat{Y}_i$.

- **Nonconformity-based, all calibration points:** every calibration compound is used, with nonconformity score $V_i = Y_i - \hat{Y}_i$ for the true label and $V_{\text{new}} = \tau - \hat{Y}_{\text{new}}$ for the (unknown) test label evaluated at the null boundary (`ConformalPredictor.get_nonconformity_scores`). This uses the full calibration set, at the cost of a symmetric rather than one-sided nonconformity notion.

Both pools are combined with each of the three weighting schemes below, giving six p-value variants per dataset/split.

3.1.4 Multiple-hypothesis correction

Applying the decision rule of Section 3.1.2 independently to every compound in a virtual library of size N controls only the per-compound Type-I error; the expected fraction of false positives in the returned shortlist can be far higher. We instead adjust the raw p-values $\{p_j\}_{j=1}^N$ with the Benjamini–Hochberg (BH) procedure [Benjamini and Hochberg, 1995] to control the FDR at level q under the (weighted-)exchangeability guarantee of Section 3.1.2, and additionally provide Benjamini–Yekutieli (BY, valid under arbitrary dependence; Benjamini and Yekutieli, 2001), Holm [Holm, 1979], and Hochberg [Hochberg, 1988] as more conservative alternatives (`ConformalPredictor.p_adjust`). A compound is selected if its adjusted p-value is $\leq q$.

3.1.5 Distinct relation to conformal outlier testing

SPDT should be distinguished from conformal p-values for outlier / novelty detection, e.g. Bates et al. [2023]. Both use the calibration-then-test paradigm of conformal inference to produce non-parametric, FDR-controllable p-values, but they differ in target and construction:

1. **Target of inference.** Bates et al. [2023] address unlabelled, unsupervised out-of-distribution testing: a model-agnostic conformity score over $\mathcal{X}$ flags structural novelty. SPDT is a supervised, directed test that thresholds the continuous response space $\mathcal{Y}$, using $\mathcal{X}$ only indirectly, through the model f and (in the weighted variants) through covariate-shift correction.
2. **Null construction.** Standard conformal regression relies on a global error distribution $|Y - \hat{Y}|$, which can be over- or under-conservative under heteroskedasticity. SPDT discards the above-threshold subpopulation entirely during calibration, so the test statistic is evaluated strictly against the model’s empirical behaviour (local bias and variance) on valid (sub-threshold) structures.

3.2 Covariate shift and importance-weighted calibration

The validity argument in Section 3.1.2 requires $(\hat{Y}_{\text{new}}, \mathcal{P}_0)$ to be (weighted-)exchangeable. When the test compound is drawn from a different region of chemical space than the calibration set — the expected situation under scaffold- or cluster-based splitting, and the realistic situation in prospective virtual screening — this assumption fails, because the calibration compounds nearest to X_{new} in structure are systematically over- or under-represented relative to a uniformly exchangeable draw. Following the weighted-conformal framework of Tibshirani et al. [2019], we correct for this by replacing the uniform calibration weights in Eq. (3.1.2) with importance weights w_i that approximate the covariate density ratio between the target (test) and calibration (control) distributions. Because the true density ratio over 2048-bit Morgan fingerprints is intractable, we approximate it using the distribution of mean k -nearest-neighbour (Tanimoto) distances from each calibration compound to the target set, and implement three schemes of increasing locality (`src/conformal_predictor.py`):

(i) Entropy balancing. Let X_i^{ctrl} be the mean 5-NN Tanimoto distance from calibration compound i to the (sub-sampled) target set, and X_j^{tgt} the corresponding mean 5-NN distance of target compound j to the rest of the target set. We solve

$$\max_{w \geq 0} - \sum_i w_i \log w_i \quad \text{s.t.} \quad \frac{1}{n} \sum_i w_i = 1, \quad \text{KS}(\{X_i^{\text{ctrl}}\}, w; \{X_j^{\text{tgt}}\}) \leq \delta,$$

where $\text{KS}(\cdot, \cdot)$ is the weighted Kolmogorov–Smirnov distance between the (re-weighted) empirical CDFs of the control and target neighbour-distance distributions, and δ (`ks_bound`) is a user-set tolerance. This is a maximum-entropy formulation: among all weight vectors that bring the two distributions within δ in KS distance, it selects the one closest to uniform, minimising variance inflation. If the unweighted KS distance is already $\leq \delta$, weights are left uniform. The optimisation is solved with SLSQP (`scipy.optimize.minimize`), with calibration sets larger than 5000 sub-sampled for tractability and the resulting weights interpolated back onto the full set by neighbour-distance rank.

(ii) Direct KS minimisation. Rather than bounding the KS distance and maximising entropy, this variant directly minimises the KS distance subject to a floor on the effective sample size (ESS), $\text{ESS}(w) = (\sum_i w_i)^2 / \sum_i w_i^2 \geq n_{\text{eff}}$ (default $n_{\text{eff}} = 30$, set to 100 in all experiments in Section 5). This is more aggressive than entropy balancing — it will use all of the available ESS budget to reduce distributional mismatch — and correspondingly more prone to variance inflation from a small number of highly up-weighted calibration compounds.

(iii) Local k-NN (Tanimoto) weighting. Rather than a single global weight vector, this scheme assigns *per-test-compound* weights: for each target compound j , its k nearest calibration neighbours (Tanimoto distance on Morgan fingerprints, $k = \text{num_nn}$, -1 meaning "all") receive weight $\exp(-\gamma \cdot d_{\text{Tanimoto}})$, renormalised to sum to k ; all other calibration compounds receive weight zero for that test point. The bandwidth γ interpolates between a uniform local average ($\gamma = 0$, $\text{ESS} = k$) and single-nearest-neighbour weighting ($\gamma \rightarrow \infty$, $\text{ESS} \rightarrow 1$). Rather than fixing γ by hand, we solve for the value that yields a target mean effective sample size n_{eff} across all target compounds via Brent root-finding on the (monotone) map $\gamma \mapsto \overline{\text{ESS}}(\gamma)$ (`ConformalPredictor._set_gamma_by_ess`).

All three schemes reduce to the unweighted SPDT of Section 3.1 when `weighted=False`, which we use throughout as the baseline against which the effect of covariate-shift correction is measured.

4 Data and Experimental Setup

We evaluate the framework in three settings of increasing control: (a) three real single-endpoint ADMET benchmarks under three splitting strategies, (b) a 100-assay ChEMBL bioactivity benchmark reused from Schiebroek et al. [2026], and (c) a synthetic scaffold-enumeration experiment in which the covariate shift is fully controlled by construction.

4.1 Splitting strategies

For every dataset we compare three train/validation/test partitions (`src/utls.py`), which differ only in how structurally close the calibration (validation) and target (test) compounds are to one another and to training:

- **Random split:** i.i.d. partition; calibration and test are exchangeable by construction and serve as the sanity-check baseline.

- **Scaffold split:** compounds are grouped by Bemis–Murcko scaffold and whole scaffold groups are assigned to train/val/test, so test-set scaffolds do not appear in calibration.
- **Cluster (Butina) split:** Morgan fingerprints (radius 3, 2048 bits) are clustered with the Butina algorithm at a Tanimoto distance threshold of 0.65 (0.30 for the ADMET data-prep pipeline), small clusters are merged into their nearest large-cluster centroid, and clusters are then assigned to train/val/test by descending size — the protocol of [Schiebroek et al. \[2026\]](#), intended to produce a more aggressive and more realistic structural shift than scaffold splitting alone.

4.2 ADMET single-endpoint benchmarks

We use three physicochemical/ADMET endpoints (`data/expansion_data_prep_with_splits_*.csv`): Caco-2 cell permeability (P_{app} A→B), human liver microsomal intrinsic clearance (HLM CL_{int}), and kinetic aqueous solubility (KSOL). Each endpoint is binarised at an assay-specific decision threshold τ (e.g. $\tau = 200 \mu\text{M}$ for KSOL) and modelled as binary classification; `conf_threshold = 0.5` on the predicted probability is used as the null boundary for SPDT (`run_conformal_fdr, src/train_xgboost.py`). Features are 2048-bit radius-2 Morgan fingerprints concatenated with the full RDKit descriptor set; an XGBoost classifier (`binary:logistic`, depth 6, 100 boosting rounds) is trained on the training split of each endpoint/split combination. For every endpoint/split we compute six BH-adjusted p-value variants: {unweighted, kNN-weighted, KS-weighted} $\times$ {negatives-only score-based, all-calibration nonconformity-based} (Section 3.1.3); $n_{\text{eff}} = 100$ for all weighted variants.

4.3 100-assay ChEMBL bioactivity benchmark

To obtain a large number of independent replicates of the shift-vs-no-shift comparison, we reuse the ChEMBL bioactivity benchmark of [Schiebroek et al. \[2026\]](#) (`data/data_schiebroek_2026/`): 100 target-assay pairs (pChEMBL activity thresholded at $\tau = 7$, i.e. $\text{IC}_{50}/K_i \leq 100 \text{ nM} \Rightarrow \text{active}$), each with its own random, scaffold, and Butina-cluster split, giving 300 independent train/calibrate/test experiments in total (`src/train_xgboost_schiebroek.py`). Because of the number of assays, this benchmark is used only with the unweighted SPDT baseline in the current codebase; it establishes how frequently and how severely FDR control is violated under structural splits before any covariate-shift correction is applied, and is the primary motivating result of this paper (Section 5.1).

4.4 Synthetic scaffold-swap experiment

Real chemical datasets confound covariate shift with sample size, label imbalance, and assay noise, all of which independently affect FDR control. To isolate the effect of covariate shift, `src/scaffold_experiment_1.py` constructs a fully synthetic benchmark: two scaffolds (a piperidine, "Scaffold A", and a trisubstituted-benzene/triazine, "Scaffold B") are each enumerated over an 18-member R-group library at three independently-varied substitution positions via RDKit `molzip`, producing on the order of 18^3 candidate compounds per scaffold (fewer after de-duplication and sanitisation failures). The binary label is $\text{MolLogP} \geq 7$ (computed from RDKit descriptors on the enumerated structure, with independent Gaussian noise added to each scaffold’s logP to avoid a degenerate, perfectly separable label). Scaffold A is over-represented in training and calibration and *absent from test*; Scaffold B is split 50%/10%/40% into train/calibration/test, so the test set is drawn entirely from a scaffold that is a small minority of the calibration set — a controlled analogue of a structural split, with a known ground-truth covariate shift direction. This lets

Table 1: Unweighted SPDT, 100 ChEMBL bioactivity assays (results/schiebroek_results_*.csv), nominal FDR $q = 0.2$. “Assays w/ selection” excludes assays where no compound was selected at $q = 0.2$ (no FDR is defined in that case). Mean true FDR and the fraction of well-controlled assays are computed over that subset.

Split	Mean AUROC	Mean AUPRC	Assays w/ selection	Mean true FDR	Frac. true FDR ≤ 0.2
Random	0.885	0.840	86/100	0.095	0.98
Scaffold	0.834	0.781	76/100	0.131	0.86
Butina cluster	0.748	0.689	59/100	0.358	0.37

us directly check, by construction, whether the importance weights recovered by KS- and kNN-weighting up-weight the (correct) Scaffold-B calibration compounds relative to Scaffold A. [TODO: Results from this experiment (weight distributions by scaffold, FDR curves on the all-Scaffold-B test set) have been produced interactively in the notebook but not yet exported to results/; re-run scaffold_experiment_1.py (or scaffold_experiment_2.py for the corresponding ablation) and save outputs before finalising this section.]

5 Results

5.1 Naive conformal FDR control degrades under structural splits

Table 1 summarises the unweighted SPDT baseline across the 100-assay ChEMBL benchmark (Section 4.3), at a nominal FDR level $q = 0.2$. Random splitting controls the FDR reliably: the mean true FDR across assays with at least one selected compound is close to nominal, and 98% of assays keep the true FDR at or below the target. Both structural splits degrade this substantially. Scaffold splitting raises the mean true FDR to 0.131 and drops the fraction of well-controlled assays to 86%. Butina cluster splitting — the more aggressive structural shift — is markedly worse: mean true FDR rises to 0.358, nearly double the nominal level, and only 37% of assays keep the true FDR at or below $q = 0.2$. Figure 1 shows the full distribution: under cluster splitting the interquartile range of true FDR sits well above the nominal target, indicating that the failure is systematic rather than driven by a handful of outlier assays.

Two observations motivate the weighting schemes of Section 3.2. First, the ranking of split difficulty for FDR control (random $\ll$ scaffold $\ll$ cluster) tracks the ranking of split difficulty for discrimination (AUROC), consistent with structural shift being the common cause of both. Second, the number of assays with *zero* selected compounds at $q = 0.2$ also rises sharply under cluster splitting (41/100, vs. 14/100 for random), i.e. the unweighted procedure is simultaneously under-powered on many assays and badly anti-conservative on others — exactly the symptom expected when exchangeability is violated non-uniformly across assays with different degrees of scaffold overlap between calibration and test.

5.2 Effect of importance weighting on ADMET endpoints

Table 2 reports, for each ADMET endpoint and split, the number of compounds selected at nominal FDR $q = 0.2$ and the resulting true FDR, for the unweighted, kNN-weighted, and KS-weighted score-based variants (Section 3.1.3; the nonconformity-based variants selected zero compounds in every configuration tested so far and are omitted – [TODO: diagnose why the all-calibration nonconformity pooling is currently over-conservative; likely the symmetric, two-sided nonconformity

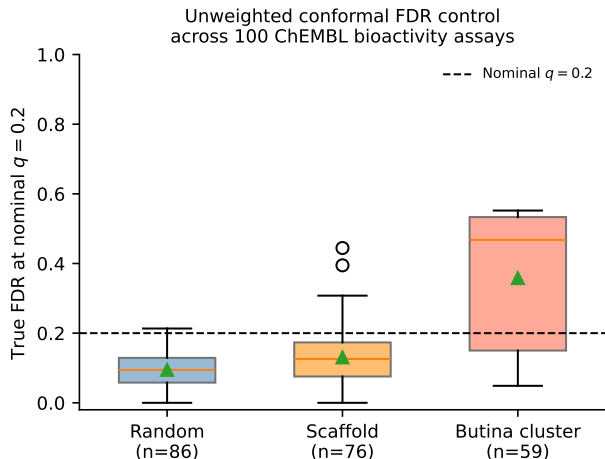


Figure 1: Distribution of true FDR at nominal $q = 0.2$ across 100 ChEMBL bioactivity assays, unweighted SPDT, by split type. Predictive performance (mean AUROC) also degrades from random to scaffold to cluster splitting, confirming that the cluster split induces a harder generalisation problem, but the FDR-control failure is disproportionately larger than the AUROC drop would suggest.

score is too permissive as a null-distribution boundary for these class-imbalanced endpoints, and needs either a one-sided nonconformity definition or a less conservative τ).

Two patterns stand out. First, unlike the 100-assay ChEMBL benchmark, all three ADMET endpoints (where any compound is selected at all) show true FDR *below* the nominal $q = 0.2$ for every method, weighted or not — these larger, more homogeneous single-endpoint datasets are evidently easier exchangeability problems than the ChEMBL benchmark’s smaller, more diverse per-assay calibration sets. Second, where covariate shift is severe enough to matter (KSOL under cluster and scaffold splitting), weighting has a consistent, visible effect on the selected set size and, for KS-weighting, on the true FDR ($0.088 \rightarrow 0.075$ under cluster split; $0.077 \rightarrow 0.065$ under scaffold split), at the cost of being more conservative (fewer compounds selected). The kNN-weighted variant is markedly more conservative still, occasionally collapsing to zero selections under severe shift (Caco-2, scaffold split) — a direct consequence of the local weighting scheme concentrating weight on very few effective calibration neighbours (Section 3.2(iii)) when the target compounds are structurally distant from all calibration compounds. HLM CLint selects zero compounds under every method and split, most likely reflecting either a poorly-separated classification boundary at the chosen threshold or too few negative calibration examples surviving the `conf_threshold = 0.5` cut; [TODO: inspect the HLM CLint score distribution and negative-calibration pool size directly (`scores_calib_neg` in the saved pickle) before drawing conclusions from this endpoint].

Figure 2 shows the full FDR-control curve (true FDR as a function of the nominal level, rather than the single value at $q = 0.2$ tabulated above) for KSOL across all three splits. Under the random split all three methods track the diagonal (nominal = true) closely at all thresholds. Under the scaffold and cluster splits the unweighted curve departs visibly from the diagonal at low nominal FDR, and both weighting schemes pull the curve back toward calibration, most consistently for KS-weighting.

Figure 3 illustrates the mechanism directly: under the KSOL scaffold split, the empirical CDF of mean 5-NN Tanimoto distance (calibration to target) sits well to the left of the corresponding target-to-target CDF before weighting (calibration compounds are, on average, structurally closer

Table 2: ADMET endpoints, nominal FDR $q = 0.2$, score-based SPDT (`results/*_fdr_metrics.csv`). “—” indicates zero compounds selected (true FDR undefined). All three splits use the same trained model per endpoint; only the calibration/test partition differs.

Endpoint	Split	Unweighted		kNN-weighted		KS-weighted	
		n_{sel}	FDR	n_{sel}	FDR	n_{sel}	FDR
Caco-2	Cluster	169	0.142	122	0.148	205	0.176
Caco-2	Scaffold	65	0.108	0	—	72	0.097
HLM CLint	Random	0	—	0	—	0	—
HLM CLint	Cluster	0	—	0	—	0	—
HLM CLint	Scaffold	0	—	0	—	0	—
KSOL	Random	117	0.111	104	0.106	106	0.094
KSOL	Cluster	34	0.088	19	0.000	40	0.075
KSOL	Scaffold	26	0.077	23	0.087	46	0.065

to other calibration compounds than target compounds are to each other, reflecting a systematic proximity gap), and both weighting schemes shift the weighted calibration CDF toward the target curve.

5.3 Synthetic scaffold-swap experiment

[TODO: Pending re-run of `scaffold_experiment_1.py` with results exported to `results/` (see Section 4.4). This section should report: (i) whether KS- and kNN-derived weights correctly up-weight Scaffold-B calibration compounds relative to Scaffold A (ground truth is known by construction here, unlike the real datasets); (ii) the resulting FDR-control curve on the all-Scaffold-B test set, weighted vs. unweighted; and (iii) whether the direction and magnitude of the weighting effect matches the real-data results in Table 2 and Figure 2, as a sanity check that the real-data effect is attributable to covariate shift specifically rather than a confound.]

6 Discussion

The central empirical finding is a dissociation between predictive performance and FDR-control validity under structural splits. AUROC drops by roughly 14 points from random to cluster splitting on the ChEMBL benchmark ($0.885 \rightarrow 0.748$), a meaningful but not catastrophic degradation by the standards of prospective virtual screening; the true FDR at nominal $q = 0.2$, by contrast, nearly doubles ($0.095 \rightarrow 0.358$), and the fraction of assays with valid control collapses from 98% to 37%. A practitioner monitoring only discrimination metrics would substantially under-estimate how unreliable a naive FDR-controlled shortlist becomes under exactly the kind of structural extrapolation that virtual screening exists to perform. This is the practical case for covariate-shift-aware calibration, independent of any improvement (or lack thereof) in the underlying predictive model.

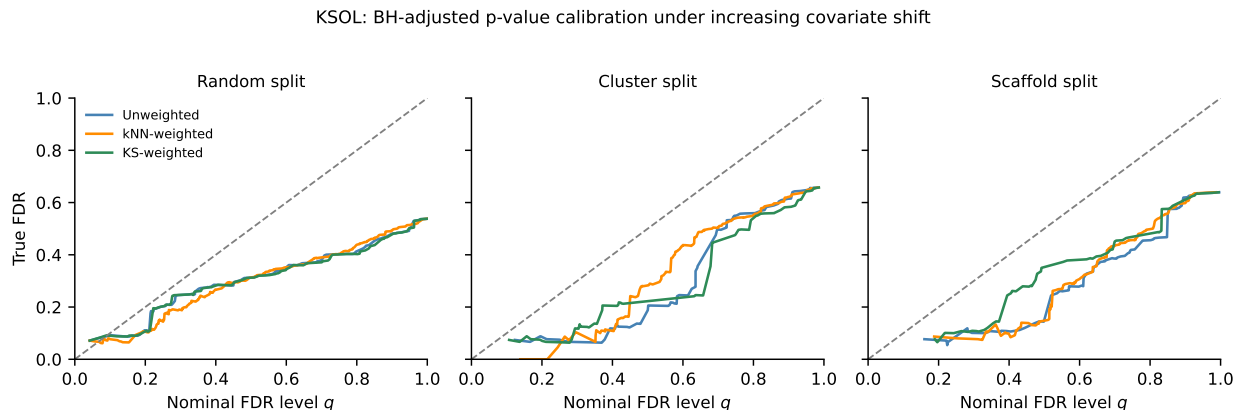


Figure 2: KSOL: true FDR vs. nominal FDR level q , unweighted vs. kNN-weighted vs. KS-weighted SPDT, across random, cluster, and scaffold splits. The diagonal is the target (true FDR = nominal FDR).

The weighting schemes evaluated here provide partial, not complete, correction, and exhibit a clear conservativeness/power trade-off: KS-weighting consistently reduces true FDR at some cost in selected-set size, while local kNN-weighting is considerably more aggressive about that trade-off and can collapse to no selections under severe shift. This matches the theoretical expectation that importance weighting trades type-I error control for reduced effective sample size, and suggests KS-weighting (which optimises a single global weight vector against an explicit ESS floor) as the more practically useful default of the two importance-weighting schemes, with entropy balancing (Section 3.2(i), not yet included in the tabulated comparison — [TODO: add entropy-balancing columns to Table 2; the predictor class supports it but `run_conformal_fdr` in `train_xgboost.py` does not currently call it]) as a potentially less variance-inflating alternative worth comparing directly.

The ChEMBL benchmark’s zero-selection assays (41/100 under cluster splitting) deserve more attention than a single aggregate true-FDR number gives them: an assay that never crosses the selection threshold is not a failure of *validity* (no false discoveries are possible in an empty set) but is a failure of *power*, and a full account of the weighting schemes’ effect on this benchmark requires re-running it with weighting enabled and reporting the joint distribution of power and validity, not validity alone. [TODO: This is the single highest-priority next experiment: currently `train_xgboost_schiebroek.py` only computes the unweighted baseline; extending it to also compute KS-/kNN-weighted p-values per assay, as already done for the ADMET endpoints in `train_xgboost.py`, would let Table 1 and Table 2 be compared on equal footing.]

Limitations. All weighting schemes here rely on nearest-neighbour Tanimoto distance in Morgan-fingerprint space as a proxy for the true covariate density ratio; this is a reasonable but crude surrogate, and its adequacy has not been directly validated against a setting with a known ground-truth density ratio other than the synthetic scaffold-swap experiment (Section 4.4, results pending). The single-endpoint ADMET results (Table 2) are each a single train/calibrate/test realisation per split, with no resampling or confidence interval on the reported true FDR; the 100-assay ChEMBL benchmark is the more statistically reliable evidence for this reason, and future revisions of this paper should either bootstrap the ADMET splits or otherwise report uncertainty on Table 2’s single-realisation FDR estimates.

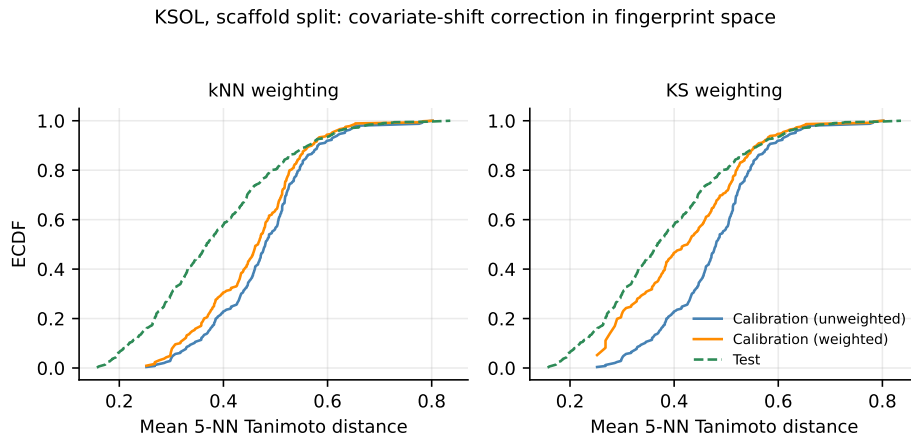


Figure 3: KSOL, scaffold split: ECDF of mean 5-NN Tanimoto distance for the calibration set before/after importance weighting, and for the target (test) set, for kNN-weighting and KS-weighting. Bringing the weighted calibration curve closer to the target curve is the explicit optimisation objective of both schemes (Section 3.2).

7 Conclusion

We formalised molecular property screening as a large-scale, one-sided hypothesis-testing problem and introduced a conformal FDR-control procedure (SPDT) calibrated against the model’s predictions on a labelled sub-threshold subpopulation. We showed, on a 100-assay ChEMBL bioactivity benchmark, that naive (unweighted) conformal FDR control is reliable under random splitting but degrades sharply under scaffold- and especially cluster-based splitting — the splitting regimes that best approximate genuine prospective virtual screening. We introduced three importance-weighting schemes that use molecular-fingerprint neighbour-distance distributions to approximate the covariate density ratio between calibration and target compounds, and showed that at least one of them (KS-distance minimisation) measurably reduces true FDR under structural shift on real ADMET data, at a cost in selected-set size. Completing the weighted evaluation of the 100-assay benchmark and the synthetic scaffold-swap experiment (Sections 5.1, 4.4) are the immediate next steps toward a full account of when, and by how much, covariate-shift-aware conformal calibration is worth its cost in statistical power.

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